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Course: **Mathematics**

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1 Basic topology

1.1 Metric spaces

Definition 1.1. A set X, whose elements we shall call *points*, is said to be *a metric space* if with any two points p and q of X there is associated a real number $d(p,q)$, called the *distance* from p to q, such that

(a) $d(p,q) > 0$ if $p \neq q$ and $d(p,p) = 0$,

(b) $d(p,q) = d(q,p)$,

(c) $d(p,q) \leq d(p,r) + d(r,q)$, for $\forall r \in X$.

Any function with these three properties is called a *distance function*, or *a metric*.

Example 1.2 (Metric spaces)**.** The following are examples of the metric spaces:

1. the set of real numbers $\mathbb{R}$ with a metric $d(p,q) = |p-q|$,
2. a real plane $\mathbb{R}^2$ with a metric $d(\mathbf{p},\mathbf{q}) = \sqrt{(p_1-q_1)^2 + (p_2-q_2)^2} := \|\mathbf{p}-\mathbf{q}\|$ (Eucledian distance),
3. a real plane $\mathbb{R}^2$ with a metric $d(\mathbf{p},\mathbf{q}) = |p_1-q_1| + |p_2-q_2|$ (Manhattan distance),
4. the set of probability distributions defined on the same measurable space with a metric $d(P,Q) = \frac{1}{\sqrt{2}}\left(\int\left(\sqrt{p(x)} - \sqrt{q(x)}\right)^2 dx\right)^{1/2}$ (Hellinger distance).

It is important to observe that every subset Y of a metric space X is a metric space in its own right, with the same distance function. Thus, every subset of a Euclidean space is a metric space.

Definition 1.3. By the *segment* (a,b) we mean the set of all real numbers x such that $a < x < b$. By the *interval* $[a,b]$ we mean the set of all real numbers x such that $a \leq x \leq b$.

If $a_i < b_i$ for $i = 1,\ldots,k$, the set of all points $\mathbf{x} = (x_1,\ldots,x_k)$ in $\mathbb{R}^k$ whose coordinates satisfy the inequalities $a_i \leq x_i \leq b_i$ $(1 \leq i \leq k)$ is called a *k-cell*. Thus, a 1-cell is an interval, a 2-cell is a rectangle, etc.

If $\mathbf{x} \in \mathbb{R}^k$ and $r > 0$, the *open* (or *closed*) *ball* B with center at $\mathbf{x}$ and radius r is defined to be the set of all $\mathbf{y} \in \mathbb{R}^k$ such that $\|\mathbf{y}-\mathbf{x}\| < r$ (or $\|\mathbf{y}-\mathbf{x}\| \leq r$).

We call a set $E \subset \mathbb{R}^k$ *convex* if

$$\lambda\mathbf{x} + (1-\lambda)\mathbf{y} \in E$$

whenever $\mathbf{x} \in E$, $\mathbf{y} \in E$, and $0 < \lambda < 1$. For example, balls are convex. It is also easy to see that k-cells are convex.

Definition 1.4. Let X be a metric space. All points and sets mentioned below are understood to be elements and subsets of X.

(a) *A neighborhood* of a point p is a set $N_r(p)$ consisting of all points q such that $d(p,q) < r$. The number r is called the *radius* of $N_r(p)$.

(b) A point p is a *limit point* of the set E if *every* neighborhood of p contains a point $q \neq p$ such that $q \in E$. Example: take a set $A := (0,1)$. Point 0 is a limit point, because any open interval, say $(-\varepsilon,\varepsilon)$, intersects A.

(c) If $p \in E$ and p is not a limit point of E, then p is called an *isolated point* of E. Example: take a set $A = \{n^{-1} : n \in \mathbb{N}\}$. Each element is an isolated point because you can take a small interval around n^{-1} that avoids the other fractions in the set.

(d) E is *closed* if every limit point of E is a point of E. Example: take $A = [0,1]$. Both 0 and 1 are limit points and both belong to the set A. A set $B = (0,1]$ is not closed because a limit point 0 does not belong to the set.

(e) A point p is an *interior point* of E if there is a neighborhood $N_r(p)$ of p such that $N \subset E$. Example: take a set $A = (0,1)$. A point 0.5 is an interior point because there is a neighborhood around it, say, $N_{0.1}(0.5)$ that belongs to the set A; if $N_{0.1}(0.5) = (0.4,0.6) := B$, we have $B \subset A$. On the other hand, if $C = [0.5,1]$, 0.5 is not an interior point of C, because there is no neighborhood around it that is a subset of C; some points of that neighborhood are outside of C.

(f) E is *open* if every point of E is an interior point of E.

(g) The *complement* of E (denoted by E^c) is the set of all points $p \in X$ such that $p \notin E$.

(h) E is *perfect* if E is closed and if every point of E is a limit point of E. Example: take $A = [0, 1]$, which is closed with all points being limit points, so it is perfect. On the other hand, $B = [0, 1] \cup \{3\}$ is not perfect because it contains a point 3, which is not a limit point (it is an isolated point).

(i) E is *bounded* if there is a real number M and a point $q \in X$ such that $d(p, q) < M$ for $\forall p \in E$.

(j) E is *dense in* X if every point of X is a limit point of E, or a point of E (or both).

Let us note that in $\mathbb{R}^1$ neighborhoods are segments, whereas in $\mathbb{R}^2$ neighborhoods are interiors of circles.

Theorem 1.5. *Every neighborhood is an open set.*

Proof. Consider neighborhood $E = N_r(p)$, and let q be any point of E. Then there is a positive real number h such that

$$d(p, q) = r - h.$$

For all points s such that $d(q, s) < h$, we have then

$$d(p, s) \leq d(p, q) + d(q, s) < r - h + h = r,$$

so that $s \in E$. Thus, q is an interior point of E. □

Theorem 1.6. *If p is a limit point of a set E, then every neighborhood of p contains infinitely many points of E.*

Proof. Suppose there is a neighborhood N of p which contains only a finite number of points of E. Let $q_1, \ldots, q_n$ be those points of $N \cap E$, which are distinct from p, and put

$$r = \min_{1 \leq m \leq n} d(p, q_m)$$

The minimum of a finite set of positive numbers is clearly positive, so that $r > 0$.

The neighborhood $N_r(p)$ contains no point q of E such that $q \neq p$, so that p is not a limit point of E. This contradiction established the theorem. □

Corollary 1.7. *A finite point set has no limit points.*

Theorem 1.8. *A set E is open if and only if its complement is closed.*

1.2 Compact sets

Definition 1.9. By an *open cover* of a set E in a metric space X we mean a collection $\{G_\alpha\}$ of open subsets of X such that $E \subset \cup_\alpha G_\alpha$.

Definition 1.10. A subset K of a metric space X is said to be *compact* if every open cover of K contains a *finite subcover*. More explicitly, the requirement is that if $\{G_\alpha\}$ is an open cover of K, then there are finitely many indices $\alpha_1, \ldots, \alpha_n$ such that

$$K \subset G_{\alpha_1} \cup \ldots \cup G_{\alpha_n}.$$

Corollary 1.11. *A set E is compact if it is both closed and bounded.*

1.3 Functions

Definition 1.12. Consider two sets A and B, whose elements may be any objects whatsoever, and suppose that with each element x of A there is associated, in some manner, an element of B, which we denote by $f(x)$. Then f is said to be a *function* from A to B (or a *mapping* from A into B). The set A is called the *domain* of f (we also say f is defined on A), and the elements $f(x)$ are called the *values* of f. The set of *all* values of f is called the *range* of f.

Definition 1.13. If for every $y \in B$ there is at most one $x \in A : f(x) = y$, the function f is said to be a 1-1 (*one-to-one*) mapping of A into B. This may also be expressed as follows: f is a 1-1 mapping of A into B provided that $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$, $x_1 \in A$, $x_2 \in A$.

Definition 1.14. Let A and B be two sets and let f be a mapping of A into B. If $f(A) = B$, we say that f maps A *onto* B. If, additionally, f is 1-1, then f is *one-to-one and onto (bijection).*

Definition 1.15. If there exists a 1-1 mapping of A *onto* B, we say that A and B can be put in 1-1 *correspondence*, or that A and B have the same *cardinal number*, or, briefly, that A and B are *equivalent*, and we write $A \sim B$.